Tira online E- Commerce Database Management System

CHAPTER – 1

1.1INTRODUCTION

Tira is a modern online beauty and personal care e-commerce platform that offers a wide range of skincare, makeup, haircare, fragrances, wellness, and grooming products from popular national and international brands. The platform provides customers with a convenient and secure shopping experience through features such as product browsing, advanced search, user registration, shopping cart, online payment, order tracking, and customer reviews.

In this project, a Database Management System (DBMS) is designed to efficiently manage all the information related to the Tira online shopping platform. The database stores customer details, product information, brand details, categories, orders, payments, delivery information, inventory, and customer feedback. By organizing data into well-structured tables and maintaining relationships between them, the system ensures data accuracy, consistency, and quick retrieval of information.

The project demonstrates how a relational database supports the daily operations of an e-commerce platform while reducing redundancy, improving security, and providing better data management for both customers and administrators.

1.2PROBLEM STATEMENT

Managing customer, product, order, and payment data manually is time-consuming and may lead to errors and data redundancy. This project develops a **Database Management System** for the **Tira Online E-Commerce Platform** to store and manage data efficiently, ensuring accuracy, security, and faster transactions.

1.3OBJECTIVE

"The main objective of this project is to design and develop a Database Management System for the Tira Online E-Commerce platform. The database is designed to store and manage customer details, product information, brand details, orders, payments, inventory, and delivery records in an organized manner. It helps reduce data redundancy, maintain data accuracy and integrity, and provide secure storage of information. The system also enables quick retrieval and updating of data, making business operations more efficient. Overall, the objective is to build a reliable and efficient database that supports the smooth functioning of the Tira e-commerce platform and provides better management of business data."

1.4Key Objectives

- To design an efficient relational database for the Tira Online E-Commerce platform.
- To securely store and manage customer information.
- To maintain product, brand, and category details.
- To manage inventory (stock) and update it automatically after each purchase.
- To process and maintain customer order records.
- To record and manage payment details securely.
- To track order delivery status.
- To reduce data redundancy through proper database design.
- To maintain data accuracy, consistency, and integrity.
- To enable fast retrieval and updating of information using SQL.
- To improve overall business efficiency and provide a better customer experience.

1.5BUSINESS REQUIREMENTS

- Manage customer registration and login details.
- Store product and brand information.
- Maintain product stock (inventory).
- Process customer orders and payments.
- Track order delivery status.
- Generate reports for customers, products, and orders.

1.6STAKEHOLDERS

- Customers
- Administrator
- Sellers/Brands
- Delivery Partners
- Payment Gateway
- System Administrator

1.7FUNCTIONAL REQUIREMENTS

- Customer registration and login.

- Browse and search products.
- Add products to the shopping cart.
- Place orders and make payments.
- Track order status.
- Manage product, customer, and order records.
- Update inventory after each purchase.

1.8NON-FUNCTIONAL REQUIREMENTS

- The system should provide secure data storage.
- The system should have a user-friendly interface.
- The system should process data quickly.
- The database should ensure data accuracy and reliability

1.9MOTIVATION OF THE PROJECT

The motivation behind this project is to understand how a **Database Management System (DBMS)** is used in a real-world e-commerce platform like **Tira**. As online shopping continues to grow, businesses need an efficient system to manage large amounts of customer, product, order, payment, and inventory data. Managing this information manually can lead to errors, duplicate data, and delays.

This project aims to design a well-structured database that stores and organizes data securely while reducing data redundancy and improving accuracy. It also provides practical knowledge of database design, table relationships, SQL operations, and data management. Overall, the project helps improve business efficiency, supports faster data retrieval, and demonstrates how a DBMS plays an important role in the successful operation of an e-commerce platform.